

Maternity Module API Documentation

Healthcare Management System

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1. Introduction

The Maternity Module API provides comprehensive endpoints for managing maternal healthcare data throughout the pregnancy, delivery, and postnatal care continuum. This documentation covers all available endpoints, data models, and configuration options for the maternity module.

Note: All endpoints require authentication and proper authorization to access patient data.

2. API Overview

2.1 Base URL

```
https://api.yourdomain.com/api/maternity
```

2.2 Authentication

All endpoints require Bearer token authentication. Include the token in the Authorization header:

```
Authorization: Bearer <your-token>
```

2.3 Response Format

All API responses follow a standard format:

```
{ "status": true, "msg": "Success message", "queryresult": { ... }, //  
or "data" depending on endpoint "totalCount": 100 // For paginated  
responses }
```

3. First Stage Labour

Management of first stage labour records including patient monitoring, cervical dilation tracking, and obstetric complications.

3.1 Endpoints

POST /first-stage-labour/create

Description: Create a new first stage labour record

Authentication: Required

Request Body:

```
{ "patient": "patientId", "dateExamined": "2025-01-09",  
  "inducedLabour": false, "phase": "Active", "duration": "4 hours",  
  "contraction": "Regular", "selectDrug": "Oxytocin",  
  "palpationExamination": "Normal", "obstetricComplication":  
  "None", "noOfVE": "2", "cervicalDilation": 6, "foetalHeartRate":  
  "140 bpm", "lastMenstrualPeriod": "2024-04-01",  
  "expectedDayOfDelivery": "2025-01-10", "notes": "Patient stable",  
  "comments": "Progressing well" }
```

GET /first-stage-labour/getall

Description: Retrieve all first stage labour records

Query Parameters:

Parameter	Type	Required	Description
limit	number	No	Number of records to return
skip	number	No	Number of records to skip
searchText	string	No	Search term for filtering

GET /first-stage-labour/get/:id

Description: Get a specific first stage labour record by ID

URL Parameters:

- `id` - The record ID

GET /first-stage-labour/patient/:patientId

Description: Get all first stage labour records for a specific patient

URL Parameters:

- `patientId` - The patient ID

PUT `/first-stage-labour/update/:id`

Description: Update an existing first stage labour record

URL Parameters:

- `id` - The record ID to update

GET `/first-stage-labour/paginated`

Description: Get paginated first stage labour records

Query Parameters:

Parameter	Type	Default	Description
page	number	1	Page number
limit	number	10	Records per page
searchText	string	-	Search filter

GET /first-stage-labour/aggregate

Description: Get aggregated statistics for first stage labour data

Query Parameters:

Parameter	Type	Description
startDate	date	Start date for aggregation
endDate	date	End date for aggregation
groupBy	string	Group by: day, month, or year

GET /first-stage-labour/count

Description: Get total count of first stage labour records

3.2 Data Model

Field	Type	Required	Description
dateExamined	Date	Required	Date of examination
inducedLabour	Boolean	Optional	Whether labour was induced
phase	String	Required	Labour phase: "Active" or "Latent"
duration	String	Optional	Duration of labour
contraction	String	Optional	Contraction details
selectDrug	String	Enum	Selected drug (see enumerations)
obstetricComplication	String	Enum	Obstetric complications (see enumerations)
noOfVE	String	Enum	Number of vaginal examinations: "1" to "6"
cervicalDilation	Number	Optional	Cervical dilation in cm (4-10)
foetalHeartRate	String	Optional	Foetal heart rate
lastMenstrualPeriod	Date	Optional	Date of last menstrual period
expectedDayOfDelivery	Date	Optional	Expected delivery date
notes	String	Optional	Clinical notes
comments	String	Optional	Additional comments
patient	ObjectId	Optional	Reference to patient record

doctor	ObjectId	Optional	Reference to doctor/user
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4. Second Stage Labour

Management of second stage labour including delivery mode, monitoring, and complications.

4.1 Endpoints

POST /second-stage-labour/create

Description: Create a new second stage labour record

Request Body:

```
{ "patient": "patientId", "modeOfDelivery": "Spontaneous Vaginal
Delivery (SVD)", "contraction": "Strong and regular",
"deliveryDate": "2025-01-09", "deliveryTime": "14:30",
"duration": "45 minutes", "selectDrug": "Oxytocin",
"preferredDrugs": "Oxytocin", "foetalMonitoring": "Continuous",
"route": "Intravenous", "obstetricComplication": "None",
"obstetricCare": "Standard", "cervicalDilatation": 10,
"comments": "Normal delivery" }
```

GET /second-stage-labour/statistics

Description: Get delivery statistics for second stage labour

Additional Feature: This endpoint provides comprehensive delivery statistics including success rates, complications, and mode of delivery distribution.

Note: Second Stage Labour includes all standard CRUD endpoints similar to First Stage Labour, plus the statistics endpoint for delivery analytics.

4.2 Data Model

Field	Type	Required	Description
modeOfDelivery	String	Required	Mode of delivery (see enumerations)
deliveryDate	Date	Required	Date of delivery
deliveryTime	String	Optional	Time of delivery
route	String	Enum	Drug administration route
obstetricCare	String	Enum	Type of obstetric care provided
cervicalDilatation	Number	Optional	Cervical dilation (4-10 cm)

5. Third Stage Labour

Management of third stage labour including mother and newborn data, delivery details, and babies data.

5.1 Endpoints

POST /third-stage-labour/create

Description: Create a new third stage labour record

Request Body:

```
{ "patient": "patientId", "mother": { "bloodPressureSystolic":
"120", "bloodPressureDiastolic": "80", "pulse": "72",
"temperature": "36.5", "respiration": "18",
"statusAfterDelivery": "Alive" }, "newBorn": { "newBornStatus":
"Live Birth", "apgarScore1Min": 8, "apgarScore5Min": 9,
"apgarScore10Min": 10, "weightKg": "3.2" }, "delivery": {
"placenta": "Complete", "membranes": "Complete", "cord":
"Normal", "placentaWeight": "500g", "bloodLoss": "250ml",
"perinealStatus": "Intact", "typeofDelivery": "SVD",
"deliveredBy": "Dr. Smith" }, "babiesData": { "liveBirth": true,
"freshStillBirth": false, "maceratedStillBirth": false,
"asphyxia": false, "lowBirthWeight": false, "macrosomicBabies":
false, "earlyNeoNatalDeath": false, "bornBeforeArrival": false,
"preMaturity": false } }
```

GET /third-stage-labour/statistics

Description: Get comprehensive statistics for third stage labour

Returns: Mother and newborn outcome statistics, complications, and success rates

5.2 Data Model - Complex Nested Structure

Mother Data

Field	Type	Description
bloodPressureSystolic	String	Systolic blood pressure
bloodPressureDiastolic	String	Diastolic blood pressure
pulse	String	Pulse rate
temperature	String	Body temperature
respiration	String	Respiration rate
statusAfterDelivery	String	"Alive" or "Dead"

NewBorn Data

Field	Type	Description
newBornStatus	String	Status: Live Birth, Fresh Still Birth, Macerated Still Birth, Birth with Deformities
apgarScore1Min	Number	APGAR score at 1 minute (0-10)
apgarScore5Min	Number	APGAR score at 5 minutes (0-10)
apgarScore10Min	Number	APGAR score at 10 minutes (0-10)
weightKg	String	Birth weight in kilograms

Delivery Data

Field	Type	Description
placenta	String	"Complete" or "Retained"
membranes	String	"Complete" or "Incomplete"
cord	String	"Normal" or "Abnormal"
placentaWeight	String	Weight of placenta
bloodLoss	String	Estimated blood loss
perinealTear	String	Degree of tear: First, Second, Third, or Fourth
episiotomy	String	"Lateral", "Medical", or "Others"
typeofDelivery	String	Type of delivery (see enumerations)
multipleGestation	String	Twin, Triplet, or Quadruplet Delivery

6. Birth Register

Official birth registration system for recording newborn details and issuing birth certificates.

6.1 Endpoints

POST /birth-register/create

Description: Register a new birth

Request Body:

```
{ "patient": "patientId", "dateOfChildRegistration": "2025-01-09", "under1YearRegistration": true, "sex": "Male",
  "placeOfBirth": "Abuja General Hospital", "childName": {
    "lastName": "Doe", "firstName": "John", "middleName": "James" },
  "fatherFullName": { "lastName": "Doe", "firstName": "Robert",
    "middleName": "William" }, "motherFullName": { "lastName": "Doe",
    "firstName": "Jane", "middleName": "Mary" }, "motherAge": 28,
  "fathersStateOfOrigin": "Lagos", "residentialAddress": "123 Main
  Street, Abuja", "phoneNumber": "+234123456789",
  "birthCertificateIssue": "2025-01-10",
  "birthCertificateCollected": "2025-01-11" }
```

6.2 Data Model

Field	Type	Required	Description
dateOfChildRegistration	Date	Required	Date of registration
under1YearRegistration	Boolean	Optional	Whether registered under 1 year
sex	String	Required	"Male" or "Female"
childName	Object	Required	Child's name (lastName, firstName, middleName)
fatherFullName	Object	Required	Father's name (lastName, firstName, middleName)
motherFullName	Object	Required	Mother's name (lastName, firstName, middleName)
motherAge	Number	Optional	Mother's age at time of birth
birthCertificateIssue	Date	Optional	Date certificate was issued
birthCertificateCollected	Date	Optional	Date certificate was collected

7. Postnatal Care

Management of postnatal care visits, monitoring mother and newborn health, and tracking outcomes.

7.1 Endpoints

POST /postnatal-care/create

Description: Create a new postnatal care record

Request Body:

```
{ "patient": "patientId", "typeOfVisit": "New", "parity": "2",
  "motherWeeks": "2", "motherDays": "3", "newBornWeeks": "2",
  "newBornDays": "3", "sexOfChild": "Female", "kangarooMotherCare":
  "Required", "numberOfBabiesDelivered": 1, "outcomeOfVisit": {
  "visit": "Discharge", "visitFor": "Routine checkup",
  "outcomeOfVisit": "Treated", "associatedProblems": "None",
  "services": ["Blood pressure", "Vaginal Examination"],
  "neonatalComplications": [], "counselling": ["Exclusive Breast
  Feeding", "Family Planning"] } }
```

GET /postnatal-care/statistics

Description: Get postnatal care statistics

Returns: Visit outcomes, complications, and service utilization data

7.2 Data Model

Field	Type	Required	Description
typeOfVisit	String	Required	"New" or "Revisit"
sexOfChild	String	Optional	"Male" or "Female"
kangarooMotherCare	String	Optional	"Required" or "Not Required"
outcomeOfVisit	Object	Optional	Visit outcome details (see sub-model)

Outcome of Visit Sub-Model

Field	Type	Description
visit	String	"LAMA", "DOPR", "Discharge", or "Refer"
outcomeOfVisit	String	"Not Treated", "Treated", "Admitted", "Referred out", "Transportation"
services	Array	Services provided during visit
neonatalComplications	Array	List of neonatal complications
counselling	Array	Counselling topics covered

8. Mortality Register

Recording and tracking maternal, neonatal, and under-five mortality cases.

8.1 Endpoints

POST /mortality-register/create

Description: Register a mortality case

Request Body:

```
{ "name": "Patient Name", "sex": "Female", "dateOfBirth": "1995-03-15", "age": 29, "patientCardNumber": "PAT-12345", "ward": "Maternity Ward", "state": "FCT", "lga": "Abuja Municipal", "maternalMortality": true, "maternalDeath": "Post-partum haemorrhage", "other": "", "neonatalDeath": "", "Deathunderfive": "", "DeathunderfiveOther": "" }
```

GET /mortality-register/statistics

Description: Get mortality statistics

Returns: Mortality rates by category, causes of death distribution

8.2 Data Model

Field	Type	Required	Description
name	String	Required	Name of deceased
sex	String	Required	"Male" or "Female"
maternalMortality	Boolean	Optional	Whether it's a maternal death
maternalDeath	String	Enum	Cause of maternal death (see enumerations)
neonatalDeath	String	Enum	Cause of neonatal death (see enumerations)
Deathunderfive	String	Enum	Cause of under-five death (see enumerations)

9. Appendix: Enumerations

Complete list of enumeration values used across the maternity modules.

Phase Options

```
["Active", "Latent"]
```

Select Drug Options

```
["Others", "Dextrose Water", "Misoprostol", "Normal Saline",  
"Oxytocin"]
```

Obstetric Complications

```
[ "Breech Presentation", "Induction of Labour", "Preterm Labour",  
"Manual Removal of Placenta", "Post-Partum Hemorrhage (PPH)",  
"Premature Rapture of Membrane (PROM)", "Ante Partum Hemorrhage  
(APH)", "Placenta Previa", "Abruptio Placenta", "Pre - Eclampsia",  
"Eclampsia", "Maternal Death", "Pregnancy Induced Hypertension (PIH)"  
]
```

Number of Vaginal Examinations (noOfVE)

```
["1", "2", "3", "4", "5", "6"]
```

Cervical Dilation Range

```
{ "min": 4, "max": 10, "options": [4, 5, 6, 7, 8, 9, 10] }
```

Mode of Delivery

```
[ "Spontaneous Vaginal Delivery (SVD)", "C-Section (CS)", "Assisted Vaginal Delivery", "Breech Delivery", "Forceps-Assisted Delivery", "Vacuum-Assisted Delivery", "Emergency C.S" ]
```

Route of Administration

```
[ "Oral", "Intraocular", "Intraotic", "Nasal", "Sublingual", "Buccal", "Inhaled", "Enteral", "Rectal", "Vaginal", "Transdermal", "Subcutaneous", "Intramuscular", "Intravenous", "Intra-Arterial" ]
```

Obstetric Care Options

```
[ "Manual Placenta removal", "Antibiotics", "Oxytocin", "Anticonvulsant", "Blood transfusion", "Others" ]
```

Maternal Death Causes

```
[ "Post-partum haemorrhage", "Sepsis", "Obstructed labour", "Abortion", "Malaria", "Anaemia", "HIV", "Other" ]
```


Neonatal Death Causes

```
[ "Prematurity", "Neonatal Tetanus", "Continental Malformation",  
  "Others" ]
```

Under-Five Death Causes

```
[ "Malaria", "Pneumonia", "Malnutrition", "Others" ]
```

Services in Postnatal Care

```
[ "Blood pressure", "Vaginal Examination", "HB", "PCV", "Urinalysis" ]
```

Neonatal Complications

```
[ "Newborn with danger signs", "Newborn with danger Signs given first  
dose of antibiotics and referred", "Neonatal Tetanus", "Neonatal  
Jaundice" ]
```

Counselling Topics

```
[ "Maternal Nutrition", "Exclusive Breast Feeding", "Complementary  
Feeding", "Family Planning", "Female Genital Mutilation", "Infection  
Prevention", "Others" ]
```

Additional Information

Error Handling

All endpoints return standardized error responses:

```
{ "status": false, "msg": "Error description" }
```

HTTP Status Codes

Code	Description
200	Success
201	Created
400	Bad Request
401	Unauthorized
404	Not Found
500	Internal Server Error

Pagination

Paginated endpoints support the following query parameters:

- `page` - Page number (default: 1)
- `limit` - Items per page (default: 10)
- `searchText` - Search filter

Contact Information

For API support and questions, please contact the development team.

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